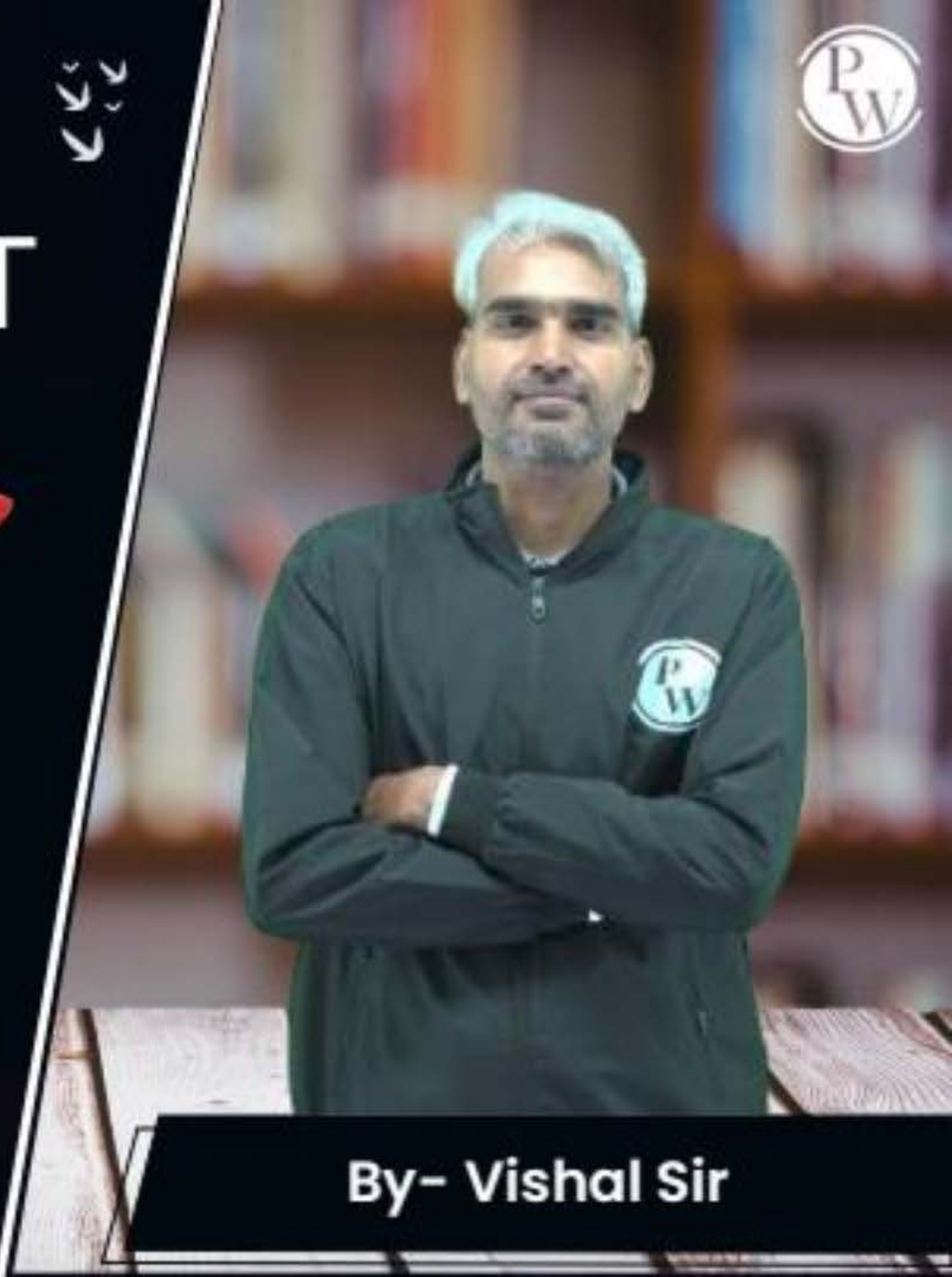
Computer Science & IT



Database Management System

Query Languages

Lecture No. 02



By- Vishal Sir

Recap of Previous Lecture



- **Topic** Query Languages
- **Topic** Relational algebra
- **Topic** Basic relational algebra operations
- **Topic** Number of super key concept

Slide

Topics to be Covered



- Topic: Derived relational algebra operations
- Topic: Join operations
- Topic: Division operation

Slide



Topic : Derived Relational Algebra operators

1. Intersection (∩)

$R \cap S = R - (R - S)$

2. Join Operations ("⋈")

3. Division Operation (÷)



Topic : Join Operations

Join operations are used to join relational tables based on some condition.

Cross Product ≡ Cartesian Product ≡ Cross Join

In Cross Join we join all tuples of first relation with all tuples of second relation



Topic : Types Join Operations

1) Inner join
   a. Theta join (Conditional join)
   b. Equi join
   c. Natural join

2) Outer join
   a. Left outer join
   b. Right outer join
   c. Full outer join



Topic : Inner join

Inner join, includes only those tuples that satisfy the matching criteria.



Topic : Outer join

In an outer join, along with tuples that satisfy the matching criteria, we also include tuples that do not match the criteria either from left hand side relation, or from right hand side relation or both.



Topic : Theta join

Conditional join

The general case of JOIN operation is called a Theta join. It is denoted by symbol θ

Theta join can use any conditions in the selection criteria.

$<, >, \leq, \geq, =$

eg. $Rel_1 \bowtie_\theta Rel_2$

θ denotes condⁿ

eg

Student

Sid	Sname
2	A
4	B

Employee

Eid	Dept-id
2	5
3	7

Student $\bowtie_{(Student.Sid \leq Employee.Eid)}$ Employee =

Student.Sid	Student.Sname	Employee.Eid	Employee.Dept-id
2	A	2	5
2	A	3	7



Topic : Equijoin

{ It is a type of theta join where condⁿ is equality condⁿ }

An equijoin is a theta join using the equality operator.
In equijoin the condition is always equality condition.

It is a type of theta join.

eg

Student

Sid	Sname
2	A
4	B

Employee

Eid	Dept-id
2	5
3	7

Student $\bowtie_{(Student.Sid = Employee.Eid)}$ Employee =

Student.Sid	Student.Sname	Employee.Eid	Employee.Dept-id
2	A	2	5

eg

Student

Sid	Sname
2	A
4	B
3	B

Enroll

Sid	Cid
2	C_1
3	C_2

Sid from both the relations will be present in the resulting relation

Student $\bowtie_{(Student.Sid = Enroll.Sid)}$ Enroll =

Student.Sid	Student.Sname	Enroll.Sid	Enroll.Cid
2	A	2	C_1
3	B	3	C_2



Topic : Natural join (⋈)

We don't need to specify the condition w.r.t. Natural Join.

W.r.t Natural Join the Condⁿ is by default Equality Condⁿ on all Common attributes

Natural join can only be performed if there is a common attribute (column) between the relations. The name and type of the attribute must be same.

Additionally, a natural join removes the duplicate columns involved in the equality comparison so only one of each compared column remains

If there is no Common attribute then, Natural join will degenerate into Cross product.

eg

Student

Sid	Sname
2	A
4	B
3	B

Enroll

Sid	Cid
2	C_1
3	C_2

Student $\bowtie$ Enroll =

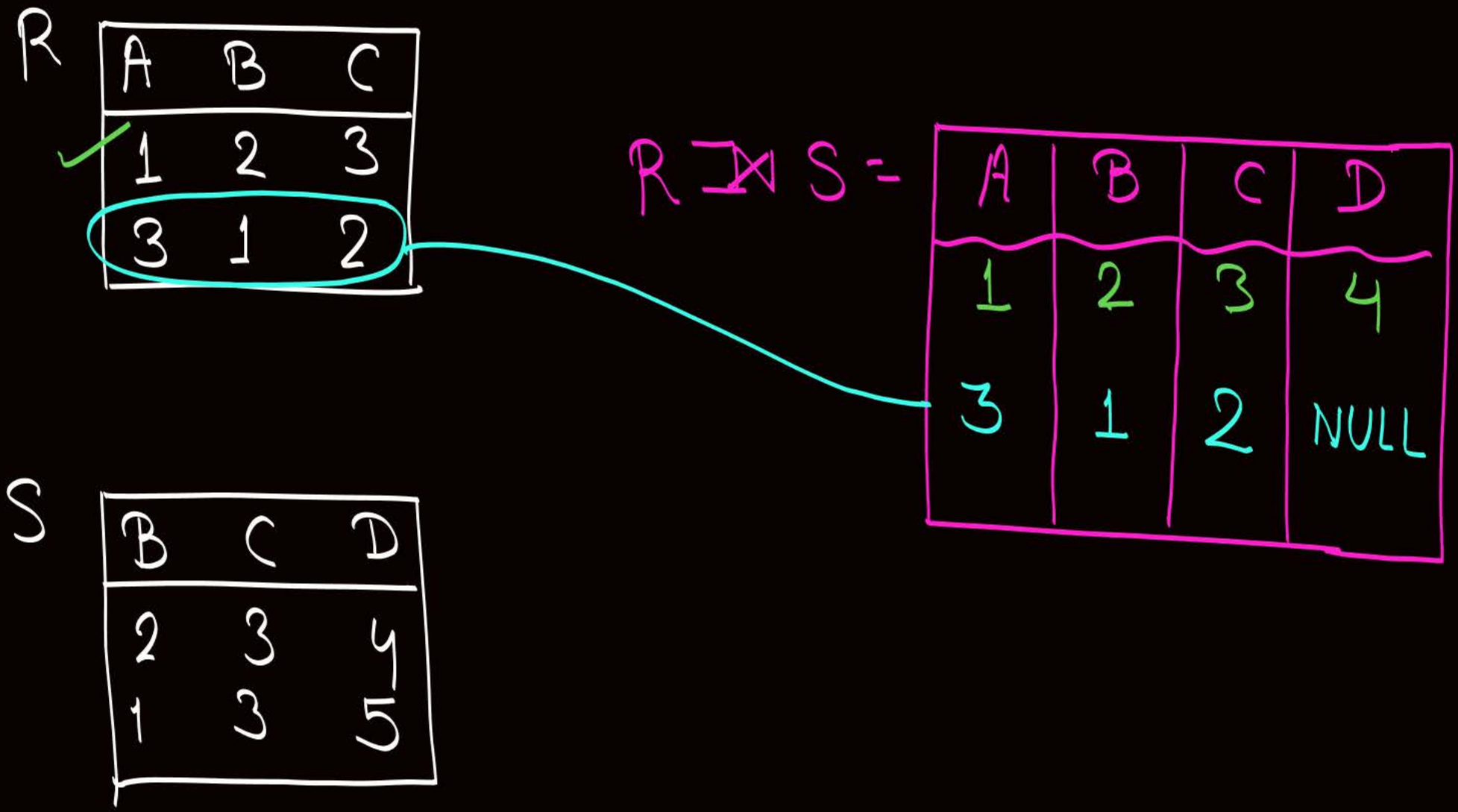
↑ We don't need to specify any Condition

Sid	Sname	Cid
2	A	C_1
3	B	C_2



Topic : Left Outer Join (⟕)

$R \overset{L}{\bowtie} S = R \bowtie S$ and tuples from the left hand side relation (i.e. Rel^n R) that failed the matching criteria (Join Cond^n)


R
A B C
1 2 3
3 1 2
S
B C D
2 3 4
1 3 5
R ⋈ S =
A B C D
1 2 3 4
3 1 2 NULL



Topic : Right Outer Join (⟖)

⟖

$R \text{ ⟖ } S$ = $R \bowtie S$, and tuples from right hand side relation (i.e 'S') that failed the matching criteria (join condn)

R

A	B	C
1	2	3
3	1	2

S

B	C	D
2	3	4
1	3	5

$R \not\bowtie S =$

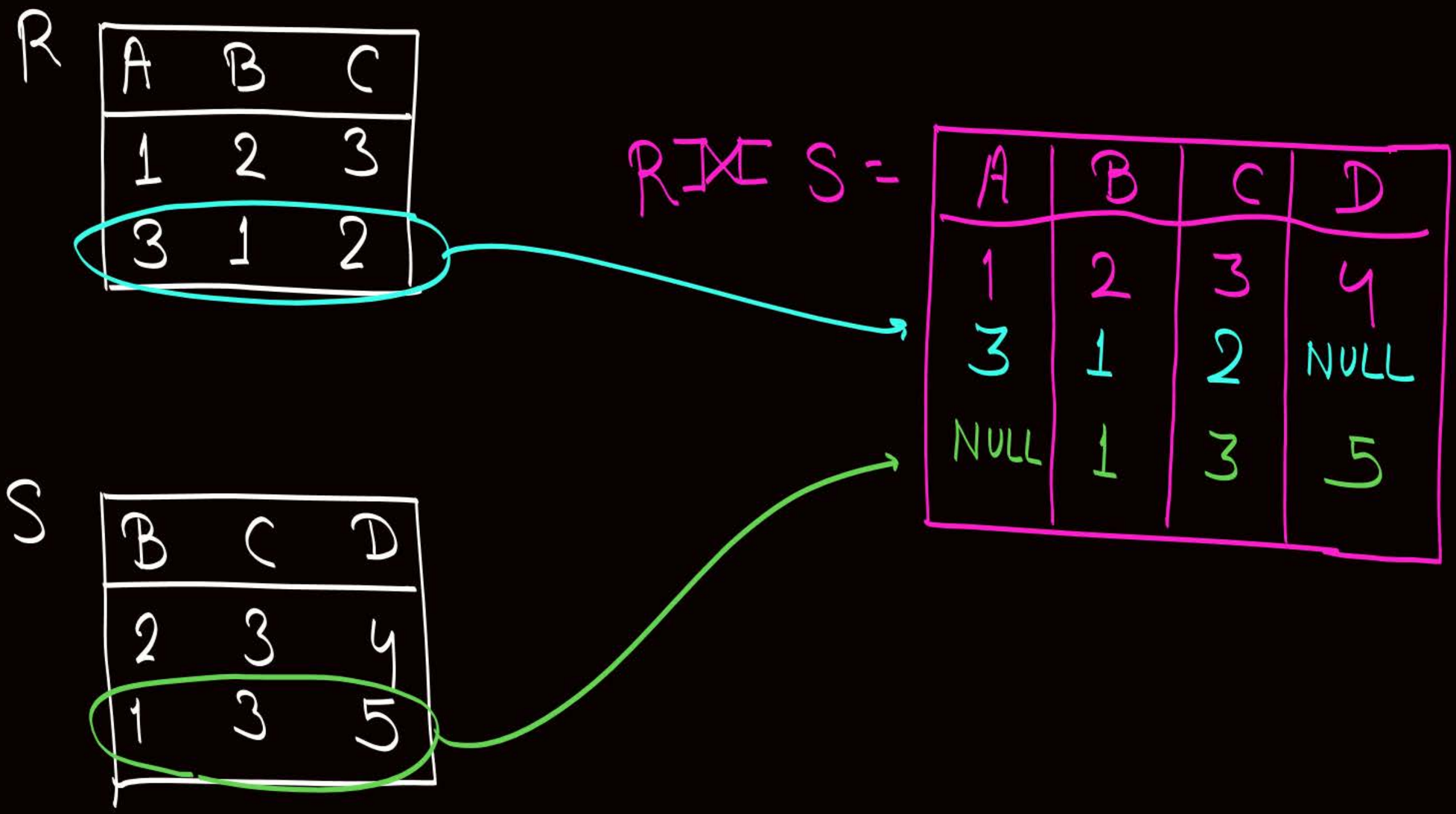
A	B	C	D
1	2	3	4
NULL	1	3	5



Topic : Full Outer Join (⟗)

⟗

$R \text{ ⟗ } S$ = $R \bowtie S$, and tuples from both left hand side as well as right hand side relations that failed the matching criteria (Join Condⁿ)


R
A B C
1 2 3
3 1 2
S
B C D
2 3 4
1 3 5
R ⋈ S =
A B C D
1 2 3 4
3 1 2 NULL
NULL 1 3 5

Q: Which among the following includes the result of other three

(a) Natural Join
(b) Left outer Join
(c) Right outer Join
(d) Full outer Join



Topic : Division (÷)

Division is a derived relational algebra opn which is derived using basic relational algebra opn.



Topic : Division (÷)

Division operation is used whenever the query is with respect to every or all.

Division operator A÷B will return those tuples from relation A which are associated to every tuple of B.

For division operator A÷B to be defined, Attributes of B must be proper subset of Attributes of A

eg: Consider the following relations

Enroll (E)

Sid	Cid
S_1	C_1
S_2	C_1
S_1	C_2
S_3	C_2
S_1	C_3
S_2	C_3

Course (C)

Cid
C_1
C_2
C_3

Query:

Retrieve Sids of the students who enrolled for all Courses ?

Answer w.r.t. given relational instance is = S_1

Soluⁿ

$$\Pi_{Sid,Cid}(Enroll) \div \Pi_{Cid}(Course)$$

$$= S_1$$

π_{Sid} (Enroll) =

Sid
S_1
S_2
S_3

It results the Sids of the Students who enrolled for some (at least one) Course.

π_{Cid} (Course)

=

Cid
C_1
C_2
C_3

It results Cids of all the courses from Course table.

$\pi_{sid}(Enroll) \times \pi_{cid}(Course)$

Sid
S_1
S_2
S_3

$\times$

Cid
C_1
C_2
C_3

$=$

Sid	Cid
S_1	C_1
S_1	C_2
S_1	C_3
S_2	C_1
S_2	C_2
S_2	C_3
S_3	C_1
S_3	C_2
S_3	C_3

It is actually a universal set in which Every student from Enroll table enroll for every Course of Course table



Topic : Division (÷)

$\pi_{Sid,Cid}(Enroll) \div \pi_{Cid}(Course)$

E C

$= [\pi_{Sid}(E) - \pi_{Sid}[(\pi_{Sid}(E) \times \pi_{Cid}(C)) - \pi_{Sid,Cid}(E)]]$

- Actual Enrollment information from Enroll table

Students who enrolled for some (at least one) Course

Sid
S_1
S_2
S_3

It will result the Sids of Students who enrolled for all Courses

Sid
S_1

Sids of the students who did not enroll for all courses

Sid
S_2
S_3

Sid	Cid
S_1	C_1
S_1	C_2
S_1	C_3
S_2	C_1
S_2	C_2
S_2	C_3
S_3	C_1
S_3	C_2
S_3	C_3

we get the universal set in which every student from Enroll table enrolled for Every Course

Sid	Cid
S_2	C_2
S_3	C_1
S_3	C_3

Sid	Cid
S_1	C_1
S_1	C_2
S_1	C_3
S_2	C_1
S_3	C_3
S_2	C_2

We will get the information that for what courses a particular student did not Enroll

This table Contain the Sids of only those students who did not enroll for all courses

eg: Consider the following relations

Enroll (E)

Sid	Cid	fee
S_1	C_1	2k
S_2	C_1	3k
S_1	C_2	2k
S_3	C_2	3K
S_1	C_3	3k
S_2	C_3	3K

Course (C)

Cid	Cname
C_1	OS
C_2	OS
C_3	DBMS

Query:

Retrieve Sids of the students who enrolled for all Courses ?

Answer w.r.t. given relational instance is = S_1

Soluⁿ

$\Pi_{Sid,Cid}(Enroll) \div \Pi_{Cid}(Course)$

$= S_1$



2 mins Summary

Topic Derived relational algebra operations

Topic Join operations

Topic Division operation

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THANK - YOU